

Tutorial - 1

A Smart Farming system Predicts Whether a Plant will grow well (1) or Poorly (0) based on daily sunlight hours and water amount in liters.

Data Set:

Sunlight hours (x_1)	Water liters (x_2)	Biog (x_3)	Target (Good) Growth = 1 / Poor = 0
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3

8

5

10

7

4

2

5

3

① Understand and Explain the data.

Inputs (x_1 & x_2)

Output (Target)

Mean of $x_1 = \mu x_1 = 6.6$

Mean of $x_2 = \mu x_2 = 3$

Variance; $\sigma^2 = \frac{1}{N} \sum (x_i - \mu)^2$

Where

N = No. of Data

x_i = Input

μ = Mean

$(x_1 - \mu x_1)^2 + (x_2 - \mu x_2)^2 + \dots$

no. of x values

$(x_1 - \mu x_1)^2 + (x_2 - \mu x_2)^2 + (x_3 - \mu x_3)^2 + (x_4 - \mu x_4)^2 + (x_5 - \mu x_5)^2$

$$= \frac{(3-6.6)^2 + (8-6.6)^2 + (5-6.6)^2 + (10-6.6)^2 + (7-6.6)^2}{5}$$

$$= 5.84$$

$$2) \frac{(x_1 - \mu_{x_1})^2 + (x_2 - \mu_{x_2})^2 + \dots}{\text{no. of values}}$$

$$= \frac{(x_{11} - \mu_{x_1})^2 + (x_{12} - \mu_{x_1})^2 + (x_{13} - \mu_{x_1})^2 + (x_{14} - \mu_{x_1})^2 + (x_{15} - \mu_{x_1})^2}{5}$$

$$= \frac{(1-3)^2 + (4-3)^2 + (2-3)^2 + (5-3)^2 + (3-3)^2}{5}$$

$$= \frac{(-2)^2 + (1)^2 + (-1)^2 + (2)^2 + (0)^2}{5}$$

$$= \frac{4+1+1+4+0}{5} = \frac{10}{5} = 2$$

Variance of $x_1 \Rightarrow \sigma^2 x_1 = 5.84$

Variance of $x_2 \Rightarrow \sigma^2 x_2 = 2$

$$\sigma^2 x_1 > \sigma^2 x_2$$

Standard deviation of

$$\sqrt{\sigma^2 x_1} = \sigma x_1 = 2.40$$

Standard deviation of

$$\sqrt{\sigma^2 x_2} = \sigma x_2 \Rightarrow 1.414$$

The factor whose variance is higher is important

$\therefore$ The sunlight hours > water liters

ML Model:

Perceptron Model:

It is a simple Neural Network Model

It is a simple decision Making Model that

can answer Yes/No.

Perceptron:-

① Inputs: x_1 & x_2 .

② Weights associated with input: w_1 & w_2

③ Bias $\rightarrow$ It gives the model flexibility (b) almost it is 1.

④ Weighted sum: $Z = w_1 x_1 + w_2 x_2 + b$

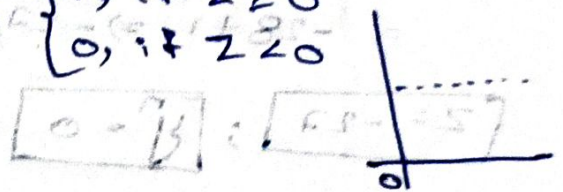
$$= \sum_i w_i x_i + b$$

⑤ Activation function $\rightarrow$ It is a mathematical (step function) function followed by ML

step function is a easiest function with

(0 & 1)

Activation function: $y^n = \begin{cases} 1, & \text{if } Z \geq 0 \\ 0, & \text{if } Z < 0 \end{cases}$



⑥ Updated weight:

$$w_i^{\text{new}} = w_i^{\text{old}} + \eta (y - y^n) x_i$$

y = Actual target/output.

$\hat{y}$ = Predicted target/output.

η = Learning Rate.

Prediction:-

① Initial:

Let; $w_1 = 0$, $w_2 = 0$, $\eta = 1$. We can take 0-1 also

Inputs $= x_1 = 3$, $x_2 = 1$, $b = 1$

$Z = ?$

$$Z = w_1x_1 + w_2x_2 + b$$

$$(0)(2) + (0)(1) + 1$$

$$Z = 1, \text{ so check is } y^n = \begin{cases} 1, & \text{if } Z \geq 0 \\ 0, & \text{if } Z < 0 \end{cases}$$

(1) $y^n = 1$ but the target is "a" so we want to change the weights.

$$w_1^{\text{New}} = 0 + 1(0-1) \cdot 3 = -3$$

$$w_2^{\text{New}} = 0 + 1(0-1) \cdot 1 = -1$$

2nd case for 2nd sunlight & water inputs

Inputs: $x_1 = 8, x_2 = 4, b = 1, w_1 = -3, w_2 = -1, \eta = 1, b = 1$

$$Z = (-3)(8) + (-1)(4) + 1$$

$$= -24 - 4 + 1$$

$$= -28 + 1 = -27$$

$$\boxed{Z = -27}; \boxed{y^n = 0} \text{ check } y^n$$

But in Dataset of 2nd row the target is 1 so it do not matched.

$$w_1^{\text{New}} = (-3) + 1(1-0)8 = -3 + (1-0)8$$

$$= -3 + 8 = 5$$

$$w_2^{\text{New}} = -1 + 1(1-0)(4) = -1 + 4$$

$$w_2^{\text{New}} = -1 + 4$$

$$w_2^{\text{New}} = 3$$

$$x_1 = 5 \quad x_2 = 2 \quad b = 1 \quad w_1 = 5 \quad w_2 = 3 \quad \eta = 1$$

$$Z = w_1x_1 + w_2x_2 + b$$

$$(5)(5) + (3)(2) + 1$$

$$25 + 6 + 1 = 32$$

$$Z = 32 \quad Z > 0 \Rightarrow y^n = 1$$

$$w_1 \text{ New} = (5) + 1(0-1)(5)$$

$$w_1 \text{ New} = 5 + (-5)$$

$$\boxed{w_1 \text{ New} = 0}$$

$$w_2 \text{ New} = (3) + 1(0-1)(2)$$

$$w_2 \text{ New} = 3 - 2$$

$$\boxed{w_2 \text{ New} = 1}$$

$$x_1 = 10, x_2 = 5, b = 1, w_1 = 0, w_2 = 1, n = 1.$$

$$z = w_1 x_1 + w_2 x_2 + b.$$

$$= (0)(10) + (1)(5) + 1.$$

$$= 0 + 5 + 1$$

$$\boxed{z = 6}$$

$$z > 0.$$

$$\text{so } \boxed{y^n = 1}$$

$\therefore$ activation function = old target.